

Hybrid RAG Search System

Combining Keyword-Based Retrieval and Semantic Retrieval for Improved Information Retrieval

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Problem Statement

Traditional search systems mainly depend on **keyword matching**, which may fail when the user's query and the document use different words but have the same meaning.

Semantic search can understand the **meaning of the query** using text embeddings, but it may sometimes miss exact keywords that are important for retrieval.

Therefore, this project develops a **Hybrid RAG Search System** that combines:

- **Keyword-based retrieval** using TF-IDF
- **Semantic retrieval** using text embeddings
- **Hybrid retrieval** by combining both scores

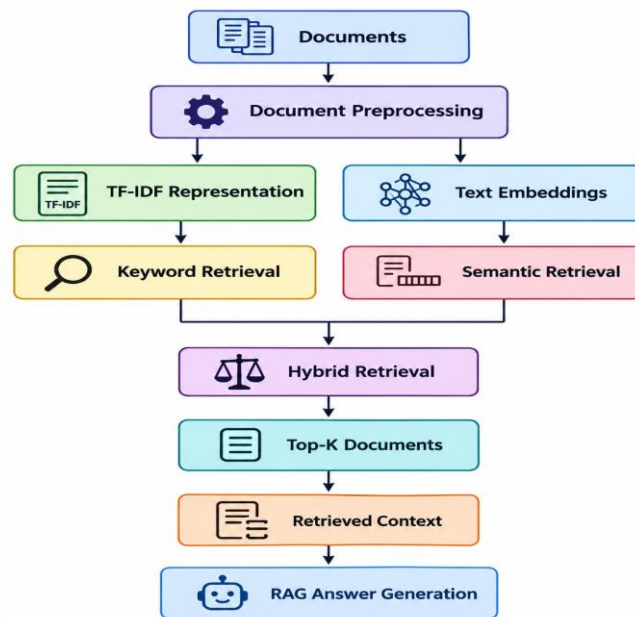
The system retrieves the most relevant documents for a given user query and provides the retrieved information as context for **Retrieval-Augmented Generation (RAG)**.

Objective

The main objectives of this project are:

1. To implement **keyword-based document retrieval** using TF-IDF.
2. To implement **semantic retrieval** using text embeddings.
3. To calculate document similarity using **cosine similarity**.
4. To combine keyword and semantic scores to perform **hybrid retrieval**.
5. To compare the performance of keyword, semantic, and hybrid retrieval methods.
6. To evaluate retrieval performance using **Precision and Recall**.
7. To demonstrate how retrieved documents can be used as context in a **RAG system**.

System Architecture



Working of the System

1. A collection of documents is provided to the system.
2. The documents are converted into **TF-IDF vectors** for keyword retrieval.
3. The documents are also converted into **embeddings** for semantic retrieval.
4. When the user enters a query, both retrieval methods search for relevant documents.
5. **Cosine similarity** is used to calculate similarity between the query and documents.
6. Keyword and semantic scores are combined to produce a **hybrid score**.
7. The documents with the highest scores are selected as the **Top-K relevant documents**.
8. These retrieved documents can be used as context for generating a RAG answer.

Methodology

The project uses three retrieval approaches.

Keyword-Based Retrieval

Keyword retrieval finds documents based on the words appearing in the query and documents.

In this project, **TF-IDF (Term Frequency–Inverse Document Frequency)** is used to represent the documents numerically.

TF-IDF gives higher importance to words that are important in a document but are less common across the entire document collection.

The similarity between the query and documents is then calculated using **cosine similarity**.

$$\text{Cosine Similarity} = \frac{A \cdot B}{\|A\| \times \|B\|}$$

where A and B are the vectors representing the query and document.

Semantic Retrieval

Semantic retrieval focuses on the **meaning of the text**, rather than only matching exact words.

In this project, the **SentenceTransformer all-MiniLM-L6-v2** model is used to convert documents and queries into numerical vectors called **embeddings**.

Documents with similar meanings have embeddings that are closer to each other.

Cosine similarity is used to compare the query embedding with document embeddings.

Hybrid Retrieval

Hybrid retrieval combines the advantages of both keyword and semantic retrieval.

The hybrid score used in the project is:

$$\text{Hybrid Score} = \alpha \times \text{Keyword Score} + (1 - \alpha) \times \text{Semantic Score}$$

In our implementation:

$$\alpha = 0.5$$

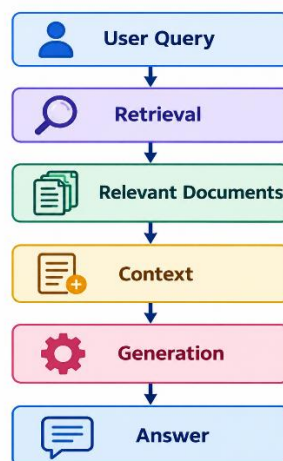
Therefore:

$$\text{Hybrid Score} = 0.5 \times \text{Keyword Score} + 0.5 \times \text{Semantic Score}$$

The documents are ranked according to their hybrid scores, and the **Top-K documents** are retrieved.

RAG Context

After retrieval, the most relevant documents are selected as context.



This allows the generation stage of RAG to use information retrieved from the document collection.

Precision and Recall

Precision and Recall are used to evaluate the quality of document retrieval.

Precision

Precision measures **how many of the retrieved documents are actually relevant**.

$$\text{Precision} = \frac{\text{Relevant Retrieved Documents}}{\text{Total Retrieved Documents}}$$

Example

If 5 documents are retrieved and 4 of them are relevant:

$$\text{Precision} = 4 / 5 = 0.80$$

So, the precision is **80%**.

Recall

Recall measures **how many of the relevant documents available in the collection were successfully retrieved**.

$$\text{Recall} = \frac{\text{Relevant Retrieved Documents}}{\text{Total Relevant Documents}}$$

Example

If there are 5 relevant documents in total and the system retrieves 4 of them:

$$\text{Recall} = 4 / 5 = 0.80$$

So, the recall is **80%**.

Importance

- **Precision** → Measures the correctness of retrieved results.
- **Recall** → Measures how many relevant documents were successfully retrieved.

Both metrics help evaluate the effectiveness of the retrieval system.

Comparison of Retrieval Methods

Retrieval Method	Working Principle	Main Advantage	Limitation
Keyword Retrieval	Matches important words using TF-IDF	Effective for exact keyword queries	May miss documents with similar meaning but different words
Semantic Retrieval	Uses embeddings to understand meaning	Can find semantically similar information	May not give enough importance to exact keywords
Hybrid Retrieval	Combines keyword and semantic scores	Considers both exact words and meaning	Requires two retrieval processes and score combination

Query Type Comparison

Keyword-based retrieval is useful when the query contains specific terms that directly appear in the documents.

Semantic retrieval is useful when the query is expressed differently from the document but has a similar meaning.

Hybrid retrieval combines both approaches and considers both keyword matching and semantic similarity.

Advantages

The major advantages of the Hybrid RAG Search System are:

1. **Combines two retrieval techniques** – Uses both keyword and semantic retrieval.
2. **Better handling of different query types** – Can handle exact keyword queries as well as meaning-based queries.
3. **Improved document retrieval** – Combining the two scores can provide more useful results.
4. **Understands semantic meaning** – Embeddings allow the system to identify documents with similar meanings.
5. **Simple evaluation** – Precision and Recall can be used to measure retrieval performance.
6. **Useful for RAG systems** – Retrieved documents can be provided as context for answer generation.
7. **Flexible** – The weight α can be adjusted to give more importance to keyword or semantic retrieval.

Results

Sample Dataset

...	Document_ID	Document	
0	1	Machine learning allows computers to learn pat...	
1	2	Deep learning uses neural networks to solve co...	
2	3	Python is widely used for data science, machin...	
3	4	SQL is used to store, retrieve and manage data...	
4	5	Natural language processing helps computers un...	
5	6	Computer vision enables computers to understand...	
6	7	Retrieval Augmented Generation combines inform...	
7	8	Embeddings represent text as numerical vectors...	
8	9	Cosine similarity measures the similarity betw...	
9	10	A vector database stores embeddings and enable...	
10	11	Data preprocessing includes cleaning missing v...	
11	12	Classification algorithms predict categories s...	
12	13	Regression algorithms predict continuous numer...	
13	14	Random forests combine multiple decision trees...	
14	15	Large language models can generate human-like ...	

Keyword Retrieval

Document_ID	Document	Keyword_Score	
0	1	Machine learning allows computers to learn pat...	0.737843
1	3	Python is widely used for data science, machin...	0.311513
2	15	Large language models can generate human-like ...	0.133452
3	2	Deep learning uses neural networks to solve co...	0.111765
4	11	Data preprocessing includes cleaning missing v...	0.088343

Semantic Retrieval

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
 WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN

modules.json: 100%  349/349 [00:00<00:00, 21.6kB/s]
 config_sentence_transformers.json: 100%  116/116 [00:00<00:00, 5.20kB/s]
 README.md: 100%  10.5k/10.5k [00:00<00:00, 502kB/s]
 sentence_bert_config.json: 100%  53.0/53.0 [00:00<00:00, 3.58kB/s]
 config.json: 100%  612/612 [00:00<00:00, 44.3kB/s]
 model.safetensors: reconstructing file: 100%  90.9MB / 90.9MB, 8.92MB/s
 model.safetensors: downloading bytes:  85.0MB, 8.23MB/s
 Loading weights: 100%  103/103 [00:00<00:00, 2124.89it/s]
 tokenizer_config.json: 100%  350/350 [00:00<00:00, 8.82kB/s]
 vocab.txt: 100%  232k/232k [00:00<00:00, 10.2MB/s]
 tokenizer.json: 100%  466k/466k [00:00<00:00, 7.82MB/s]
 special_tokens_map.json: 100%  112/112 [00:00<00:00, 3.39kB/s]
 config.json: 100%  190/190 [00:00<00:00, 6.95kB/s]

	Document_ID	Document	Semantic_Score	
0	1	Machine learning allows computers to learn pat...	0.767540	
1	2	Deep learning uses neural networks to solve co...	0.392354	
2	6	Computer vision enables computers to understan...	0.374741	
3	12	Classification algorithms predict categories s...	0.350768	
4	5	Natural language processing helps computers un...	0.343619	

Hybrid Retrieval

	Document_ID	Document	Keyword_Score	Semantic_Score	Hybrid_Score	
0	1	Machine learning allows computers to learn pat...	0.7378	0.7675	0.7527	
1	3	Python is widely used for data science, machin...	0.3115	0.2837	0.2976	
2	2	Deep learning uses neural networks to solve co...	0.1118	0.3924	0.2521	
3	6	Computer vision enables computers to understan...	0.0000	0.3747	0.1874	
4	15	Large language models can generate human-like ...	0.1335	0.2216	0.1775	

Evaluation

Keyword Retrieval
 Retrieved: [1, 3, 15]
 Precision: 0.67
 Recall: 0.67

Semantic Retrieval
 Retrieved: [1, 12, 13]
 Precision: 0.33
 Recall: 0.33

Hybrid Retrieval
 Retrieved: [1, 3, 2]
 Precision: 0.67
 Recall: 0.67

Retrieved Context:

Machine learning allows computers to learn patterns from data.
 Python is widely used for data science, machine learning and artificial intelligence.
 Deep learning uses neural networks to solve complex problems.

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QUERY: machine learning

KEYWORD:
Document_ID Document
0 1 Machine learning allows computers to learn pat...
1 3 Python is widely used for data science, machin...

SEMANTIC:
Document_ID Document
0 1 Machine learning allows computers to learn pat...
1 12 Classification algorithms predict categories s...

HYBRID:
Document_ID Document
0 1 Machine learning allows computers to learn pat...
1 3 Python is widely used for data science, machin...
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QUERY: computers understanding human language

KEYWORD:
Document_ID Document
0 5 Natural language processing helps computers un...
1 15 Large language models can generate human-like ...

SEMANTIC:
Document_ID Document
0 5 Natural language processing helps computers un...
1 15 Large language models can generate human-like ...

HYBRID:
Document_ID Document
0 5 Natural language processing helps computers un...
1 15 Large language models can generate human-like ...
=====

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QUERY: database data storage

KEYWORD:
Document_ID Document
0 10 A vector database stores embeddings and enable...
1 1 Machine learning allows computers to learn pat...

SEMANTIC:
Document_ID Document
0 4 SQL is used to store, retrieve and manage data...
1 10 A vector database stores embeddings and enable...

HYBRID:
Document_ID Document
0 4 SQL is used to store, retrieve and manage data...
1 10 A vector database stores embeddings and enable...
=====

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Conclusion

The **Hybrid RAG Search System** was successfully implemented by combining **keyword-based retrieval and semantic retrieval**.

TF-IDF was used for keyword-based retrieval, while **SentenceTransformer embeddings** were used for semantic retrieval. **Cosine similarity** was used to measure the similarity between queries and documents.

The hybrid approach combines keyword and semantic scores to retrieve relevant documents. The system was also evaluated using **Precision and Recall** for different query types.

Overall, the project demonstrates how combining traditional keyword search with semantic search can provide a flexible retrieval mechanism for **Retrieval-Augmented Generation (RAG)** systems.